



Politecnico di Torino

Qubit Electronics

Lab 2

F. Riente

March 23, 2025

Lab submission instructions

Create a folder with the following name:

`gr<nn>_lab2`

where n corresponds to the group number.

Inside that folder, insert the .pdf with the report and any additional mat-lab/python scripts you are going to develop for processing VNA output data. *The report's length must be limited to 15 pages.*

At the end of the lab session, create a .zip archive starting from the root folder and name it

`gr<nn>_lab2.zip`

The created archive must be uploaded by one team member on Portale della Didattica, section Elaborati.

1 Measurement and Analysis of RF Devices Using Vector Network Analyzer (VNA)

The objective of this lab session is to introduce you to the measurement of unknown RF devices using a Vector Network Analyzer (VNA). You will learn to interpret the scattering parameters (S-parameters) of the device under test (DUT) to understand its functionality. In addition, you familiarize yourself with the VNA software that will be used for measurement and analysis. The final report should include detailed descriptions and analyses of the DUT based on the measured S-parameters, accompanied by relevant plots and images.

Equipment and Materials:

- Vector Network Analyzer (VNA)
- RF cables and connectors
- Device under test (DUT) - Unknown RF devices
- Computer with VNA software

The report should include: An introduction to the objectives of the lab. A brief description of the theory behind S-parameters and their significance in RF device characterization. A methodology section that details the setup, calibration, and measurement process. An analysis section where the S-parameter data is presented in plots. Each plot must be accompanied by a discussion interpreting the data and hypothesizing the function of the DUT. High-quality images of the DUT connected to the VNA setup. Remember that with the downloaded data from the VNA you can use python scripting, matlab or other scripting tool to plot the S-parameters without the need of the VNA software. A conclusion summarizing the findings and reflecting on the learning outcomes of the lab session.

Please refer to the [vna_guide.pdf](#) uploaded on the Portale della Didattica if you have doubts about VNA calibration and measurements of the scattering parameters.